

A Minimal Model for Type-1 DM Patients: Meal and Exercise Adaptation

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Abstract: Mathematical Modeling of glucose-insulin dynamics of Type-1 Diabetic Mellitus (T1DM) patients is an essential component of designing and developing Artificial Pancreas Systems. These model parameters, which exhibit significant inter-patient variability, are identified for each individual T1DM patient through standard tolerance tests. However, in addition to the inter-patient variability, each patient's model parameters vary according to the circadian rhythm. Therefore, the blood glucose response to a meal is different at breakfast when compared to lunch and dinner. In addition to this, the glucose-insulin dynamics vary when the T1DM patient exercises or during any physical activity. To account for these intra-patient variabilities and the variability due to exercise, a neuro-adaptive learning scheme is proposed in this work. The uncertainties are approximated as a product of a weight and a meaningful basis function. The model uncertainties are learned during meals and idle activity, whereas exercise learning requires an announcement from the patient and is only learned when the patient is exercising. This neuroadaptive learning scheme can prove to be of vital importance in designing model-based control laws for blood glucose regulation in Type-1 Diabetic patients.

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1. INTRODUCTION

Type-1 Diabetic Mellitus (T1DM) patients cannot regulate their blood glucose (BG) levels. They depend on regular insulin injections and have strict dietary restrictions. An Artificial Pancreas (AP) system is a closed-loop system that is designed to regulate the BG levels within 70 – 180 milligrams/decilitre (mg/dL) (considered as *safe region* see: Beck et al. (2019)). The primary components of an AP system are (i) a Continuous Glucose Monitor (CGM) that senses the glucose levels in the blood, (ii) an insulin pump that can deliver insulin subcutaneously, and (iii) a feedback control algorithm to compute the insulin required. A critical component of the AP system is the control algorithm that ensures that the BG levels are always within *safe limits*. In a typical AP system, the insulin infusion rate is the control input, and the glucose level measured by the CGM is the output. The input is calculated based on output measurement in a feedback control system. Several control schemes for AP systems have been proposed in the last decade, such as Proportional Integral Derivative (PID) control, pole placement, back-stepping, and linear quadratic regulator. However, these feedback-based techniques are *reactive*. Model Predictive Control (MPC), which is from optimal control theory, is a suitable candidate for the AP system due to the predictive nature of the method and the constraint-handling capabilities it offers (Zavitsanou et al. (2016)). A key point to note is that a mathematical model of the system is required in several advanced control techniques mentioned above. The model-based control techniques exploit this mathematical representation to deliver accurate insulin doses to regu-

late the patient's blood glucose levels. This model should accurately represent the glucose-insulin dynamics of the patient.

Mathematical models have been well-developed over the last century in most physical systems (such as aircraft and robots). However, in biomedical systems, it is quite difficult to capture the internal dynamics accurately, and it is even more difficult to model a system that can be generalized. Therefore, if an accurate model of the glucose-insulin dynamics is not included in the MPC formulation, the computed insulin command may not effectively regulate blood glucose. Several groups, such as Dalla Man et al. (2007) and Wilinska and Hovorka (2008), have developed mathematical models to describe various aspects of the internal dynamics such as meal digestion, glucose absorption, and the glucose-insulin interaction in the bloodstream. The parameters of the mathematical models are identified by subjecting the patients to standard Mixed Meal Tolerance Tests (MMT) and Oral Glucose Tolerance Tests (OGTT). Patients are given a standard amount of carbohydrates during these tests and injected with insulin subcutaneously. Blood samples are drawn in timed intervals, which carry vital information such as the glucose and insulin concentration in the blood just before and after the meal and insulin is administered. This data is then used to identify the model's parameters through standard techniques such as maximum likelihood estimation and least-squares minimization. The tolerance tests and the parameter identification process is done for individual patients due to large variation in the dynamics between patients (known as *inter-patient variability*). However, it

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has been proven that significant variability exists for the same patient (also known as *intra-patient variability*) on different days and even on different meals on the same day. Therefore, the accuracy of the mathematical model fitted for the morning meal cycle may not accurately represent the same patient's lunch or dinner meal response.

In addition to this uncertainty, the patient's internal glucose-internal dynamics also vary significantly during physical activity or exercise. Physical exercise is recommended by the American Diabetic Association (Colberg et al. (2016)). However, according to Bohn et al. (2015), around 63% of Diabetic patients do not exercise regularly because it may lead to hypoglycemia. Even though the effect of exercise on human physiology has been well studied, only a few models have been presented that accurately represent the effect of exercise on glucose and insulin dynamics of the human body. Some of the studies conducted on modeling physical activity include works done by Romeres et al. (2021) and Resalat et al. (2019).

Therefore, to address the issues of parameter uncertainty, there have been several works that have proposed the use of adaptive control strategies to improve the model in real-time (See: Hajizadeh et al. (2019)). Adaptive control uses feedback from output measurements and state estimates to evaluate the error in the model and use these errors to improve the model in real time. This work proposes a neuro-adaptive learning scheme where the uncertainty model parameter and the exercise effect are approximated as a neural network where the weights of the network are learned in real time. The novel strategy proposed here has two learning schemes: uncertainty during meals is learned as a separate weight while exercise effect is learned as a separate set of weights. This separation in the weights is introduced because the increased glucose degradation due to the effect of exercise is only temporary and does not persist throughout the day. This paper details the neuroadaptive scheme used, the weight update rule, as the basis function selection. The proposed method was tested on virtual patients from the FDA-approved T1DM simulation, which is discussed next.

2. VIRTUAL PATIENT DATA-SET

The data used for this work was collected from ten virtual T1DM patients in the FDA-approved T1DM simulator. A standardized Mixed Meal Tolerance Test was configured in the simulator, where 80 grams of a pure Carbohydrate meal was given to them along with subcutaneous rapid-acting insulin (0.2 Units/kg) at time t_0 minutes. The MMT was configured to last for three hours, during which 25 samples of intravenous blood were drawn from them. A set of ten virtual patient data were generated from the simulator. It should be noted that *the effect of exercise, however, is not captured in the T1DM simulator*. Therefore, work done by Romeres et al. (2021) has also been incorporated to simulate the effect of exercise by changing the system dynamics that represent the nominal minimal model. The changes to the minimal model are described in Section 3.2,

3. MATHEMATICAL MODELING OF SUB-SYSTEMS

In this section 3.1, the nominal model for glucose-insulin dynamics is described first. However, this model only accounts for glucose excursions due to meals. The second part of this section describes how the effect of exercise can be modeled as a variation of the model parameters.

3.1 Nominal Minimal Model

The minimal mathematical model developed by Bergman (1989) describes the glucose-insulin interaction in the body and has been well studied and used extensively. The minimal model has three components: (i) a meal model developed by Dalla Man et al. (2007), which describes how carbohydrate content in a meal appears in the bloodstream (ii) a glucoregulatory model Eq.(1), which describes the glucose-insulin interaction in the body and (iii) a subcutaneous insulin kinetics model which describes the transport of insulin from subcutaneous insulin layer to the bloodstream. The main focus of this work is the gluco-regulatory portion given in Eq. (1). It describes the interaction between the glucose (G) and insulin in the remote compartment (X). The key parameters of this model are glucose-effectiveness (p_1) and insulin sensitivity ($S_I = p_3/p_2$).

$$\begin{aligned}\dot{G}(t) &= -p_1 G(t) - X(t)G(t) + p_1 G_b + \frac{R_a(t)}{V_g} \\ \dot{X}(t) &= -p_2 X(t) + p_3 \left(\frac{I_p(t)}{V_I} - I_b \right)\end{aligned}\quad (1)$$

The parameters of the minimal model are identified through nonlinear least-squares estimation by minimizing the residual error between the 25 samples of MMT data from the virtual patients and the output from the mathematical model. However, as mentioned in the Introduction section, there are uncertainties in the mathematical model compared to the actual patient data. Therefore, this model can be considered a nominal system, with *no consideration for parameter variability and no exercise incorporated into it*.

3.2 Physical Activity/Exercise Model

Physical Activity/Exercise models have also been studied in several research projects in the past two decades. According to Romeres et al. (2021), the effect of exercise modifies the glucose dynamics in two ways: (i) insulin-independent rate of disappearance (which is given by parameter p_1 Eq.1 and (ii) insulin sensitivity parameter S_I .

$$\begin{aligned}p_{1_{ex}} &= p_1(1 + p_5 E(t)) \\ S_{I_{ex}} &= S_I(1 + p_6 E_a(t))\end{aligned}\quad (2)$$

Note that p_5 and p_6 are exercise's insulin-independent and insulin-dependent effects on the glucose disappearance rate, respectively. $E(t)$ is the exercise level intensity that varies from 0 to 1 and $E_a(t)$ is the delayed signal of exercise which is given by Eq.(3). The parameter p_4 describes the delay involved

$$\dot{E}_a(t) = -p_4 E_a(t) + p_4 E(t) \quad (3)$$

The intensity (or) level of exercise can be estimated from measurements such as heart rate (HR) elevation or oxygen

saturation levels in the blood. However, the parameters of the above exercise model are unknown; therefore, a neuro-adaptive scheme is proposed to learn the effect of exercise on the model parameters using glucose measurements from the CGM and exercise intensity level.

4. LEARNING THE PARAMETER UNCERTAINTY AND EXERCISE

In the proposed neuro-adaptive scheme, it is assumed that the plant model dynamics are known partially, and the unknown dynamics are approximately modeled as a weight and a basis function. The weight update rule is explained in detail by Tiwari and Padhi (2018). The actual plant dynamics, which can be considered as a combination of the nominal dynamics and the un-modeled dynamics ($d(\mathbf{X}) \in \mathbb{R}^9$) can be represented as in Eq. 4.

$$\dot{\mathbf{X}} = f(\mathbf{X}) + g(\mathbf{U}) + d(\mathbf{X}) \quad (4)$$

A method of correctly identifying the disturbance term as well as the derivative information was proposed by Tiwari and Padhi (2018). A brief description of the technique is given below. An observer is first given by Eq. (5).

$$\dot{\mathbf{X}}_o = f(\mathbf{X}) + g(\mathbf{U}) + \hat{d}(\mathbf{X}) + K(\mathbf{X} - \mathbf{X}_o) \quad (5)$$

In the above equation, $\mathbf{X}_o$ is the observer state vector, $\hat{d}(\mathbf{X})$ is the approximated disturbance/un-modelled dynamics and K is a tuning parameter and is positive definite matrix. The objective of neuroadaptive control is to reduce the error between the actual state and the observed state ($\dot{\mathbf{E}} = \dot{\mathbf{X}} - \dot{\mathbf{X}}_o$). This error can also be defined in terms of $d(\mathbf{X})$ and $\hat{d}(\mathbf{X})$ as: $\dot{\mathbf{E}} = d(\mathbf{X}) - \hat{d}(\mathbf{X}) - KE$. Each channel (which corresponds to a particular state) in the error dynamics can be written in terms of the disturbance/un-modelled dynamics in that channel as given in Eq. (6).

$$\begin{aligned} \dot{e}_i &= \dot{x}_i - \dot{x}_{o_i} \\ &= d_i(\mathbf{X}) - \hat{d}_i(\mathbf{X}) - k_i e_i \end{aligned} \quad (6)$$

As mentioned in the introduction, $\hat{d}_i(\mathbf{X})$ is approximated as a weight and a basis function (or a linear in weight neural network) as given below in Eq.(7).

$$\hat{d}(\mathbf{X}) = \hat{W}_i^T \phi_i(\mathbf{X}) \quad (7)$$

The true disturbance/uncertainty in dynamics $d(\mathbf{X})$ can also be approximated as a weight and bias term with some small error ϵ . This will be represented an ideal weight W_i as given in $d_i(\mathbf{X}) = W_i^T \phi(\mathbf{X}) + \epsilon_i$. In order to accurately approximate the disturbance/un-modelled dynamics, the error between the ideal weight and the approximated weight $\tilde{W}_i = W_i - \hat{W}_i$ needs to be driven to zero. Therefore, the objective of the neuroadaptive scheme should be to :

- To drive the error in the state channel e_i to zero
- To drive the error in the weights $\tilde{W}_i$ to zero for each channel
- To drive the error in the Jacobian $[(\frac{\partial d_i(\mathbf{X})}{\partial \mathbf{X}}) - (\frac{\partial \hat{d}_i(\mathbf{X})}{\partial \mathbf{X}})]$ to zero.

The Lyapunov candidate function chosen to achieve these objectives is given in Eq. (8).

$$\begin{aligned} V_i(e_i, \tilde{W}_i) &= \beta_i \frac{e_i^2}{2} + \frac{\tilde{W}_i^T \tilde{W}_i}{2\gamma_i} + \\ &\quad \left[\frac{\partial d_i(\mathbf{X})}{\partial \mathbf{X}} - \frac{\partial \hat{d}_i(\mathbf{X})}{\partial \mathbf{X}} \right]^T \\ &\quad \frac{\Theta_i}{2} \left[\frac{\partial d_i(\mathbf{X})}{\partial \mathbf{X}} - \frac{\partial \hat{d}_i(\mathbf{X})}{\partial \mathbf{X}} \right] \end{aligned} \quad (8)$$

Where $\beta_i, \gamma_i, \Theta_i > 0$ are design tuning parameters. From the above Lyapunov candidate, a weight update rule can be derived (see Tiwari and Padhi (2018) for derivation.) This weight update rule guarantees that the three above mentioned errors are driven to zero.

$$\dot{\tilde{W}}_i = \left(\frac{I_i}{\gamma_i} + \frac{\partial \phi_i(\mathbf{X})}{\partial \mathbf{X}} \Theta_i \frac{\partial \phi_i(\mathbf{X})}{\partial \mathbf{X}} \right) \quad (9)$$

4.1 Approximating Parameter Uncertainty

According to Hinshaw et al. (2013), it was shown that key parameters such as the insulin sensitivity (S_I) vary according to a diurnal pattern in Type-1 Diabetic patients and increase during lunch when compared to the S_I value during breakfast. However, it is not practical to identify the variation of parameters with multiple tolerance tests. Therefore, these parameter variations can be learned using the error between the actual state and the states obtained from propagating the mathematical model with nominal parameters. As mentioned in the previous section, the learning process is done separately according to the error in each individual state.

In this research, parameter uncertainty in the following states G, X are considered as it was shown by Hinshaw et al. (2013) that insulin sensitivity varies significantly. In addition to the insulin sensitivity, glucose effectiveness (p_1) and the glucose-insulin interaction, given by the bi-linear term, XG are also considered to be perturbed.

$$\begin{aligned} \dot{G}(t) &= -(p_1 + \Delta p_1)(G(t) - G_b) - (1 + \Delta k)X(t)G(t) \\ &\quad + \frac{R_a(t)}{V_g} \\ \dot{X}(t) &= -p_2 X(t) + p_2(S_I + \Delta S_I)(I(t) - I_b) \end{aligned} \quad (10)$$

The above-given equations can be expressed as a combination of the nominal system and unmodelled system dynamics (which are a function of the state vector $\mathbf{X}$) as follows:

$$\begin{aligned} \dot{G}(t) &= -p_1(G(t) - G_b) - X(t)G(t) \\ &\quad + \frac{R_a(t)}{V_g} + d_1(\mathbf{X}) \\ \dot{X}(t) &= -p_2 X(t) + p_2 S_I(I(t) - I_b) + d_2(\mathbf{X}) \end{aligned} \quad (11)$$

However, the terms $d_1(\mathbf{X}), d_2(\mathbf{X})$ are unknown and can be approximated as $\hat{d}_i(\mathbf{X}) = \hat{W}_i^T \phi(\mathbf{X}), i = 1, 2$. The meaningful basis functions ($\phi_i(\mathbf{X})$) for this particular set of system dynamics given in Eq.(14) are selected as follows:

$$\begin{aligned} \phi_1(\mathbf{X}) &= [G(t) - G_b, XG]^T \\ \phi_2(\mathbf{X}) &= [I(t) - I_b] \end{aligned} \quad (12)$$

The basis functions are selected as given above to be compatible with the structure of the original system dynamic equations. The weights $\hat{W}_1 \in \mathbb{R}^2, \hat{W}_2 \in \mathbb{R}^1$ are initialized

as zero and updated according to the weight-update rule given in Eq. (9).

4.2 Approximating Effects due to Exercise

As mentioned in Section 3.2, exercise impacts the insulin-dependent and insulin-independent rate of glucose decrease. However, the parameters representing the effect of exercise (p_5, p_6) are difficult to identify through standard parameter identification techniques using blood sample data. Therefore, the effects on the parameters can also be learned in real-time by approximating them as another set of weights and basis functions. The effect of exercise is represented as a variation on the true patient parameters. For example, the effect of exercise on insulin-independent glucose decay was shown as $p_{1_{ex}} = p_1(1 + p_5 E(t))$, where it is assumed that the true value of the term p_1 is known. However, in the above Section 4.1, it was mentioned that the true model parameters cannot be identified accurately and may also vary with a diurnal pattern. Therefore the true parameter may be $p_1 + \Delta p_1$. The effect of exercise on the dynamics are given as follows:

$$\begin{aligned} p_{1_{ex}} &= (p_1 + \Delta p_1)(1 + p_5 E(t)) \\ S_{I_{ex}} &= (S_I + \Delta S_I)(1 + p_6 E_a(t)) \end{aligned} \quad (13)$$

In the previous section, a method of approximating the uncertainties was presented where $\Delta p_1 \approx \hat{W}_1(1)$ and $\Delta S_I \approx \hat{W}_2$. (Note: $\hat{W}_1(1)$ is the first element in $\hat{W}_1 \in \mathbb{R}^2$)

Assuming that the uncertainties in the model are already learned and substituting approximations of the uncertainties in Eq. (13). There are still uncertain terms p_5, p_6 in the exercise model which need to be identified. The effect of exercise on the model parameters can also be approximated using the same technique. It can be given as an additional term in each state as follows:

$$\begin{aligned} \dot{G}(t) &= -p_1(G(t) - G_b) - X(t)G(t) + \frac{R_a(t)}{V_g} \\ &\quad + \hat{W}_1^T \phi_1(\mathbf{X}) + d_{1_e}(\mathbf{X}) \end{aligned}$$

$$\dot{X}(t) = -p_2 X(t) + p_2 S_I (I(t) - I_b) + \hat{W}_2^T \phi_2(\mathbf{X}) + d_{2_e}(\mathbf{X}) \quad (14)$$

Here too, the terms $d_{1_e}(\mathbf{X}), d_{2_e}(\mathbf{X})$ are unknown and can be approximated as a weight and basis function ($\hat{d}_{i_e}(\mathbf{X}) = \hat{W}_{i_e} \phi_{i_e}(\mathbf{X}), i = 1, 2$). In this case, the basis functions that are selected are a function of the nominal parameters as well as the weights ($\hat{W}_1, \hat{W}_2$).

$$\begin{aligned} \phi_{1_e}(\mathbf{X}) &= [(p_1 + \hat{W}_1)E(t)(G(t) - G_b)] \\ \phi_{2_e}(\mathbf{X}) &= [p_2(S_I + \hat{W}_2)(I(t) - I_b)] \end{aligned} \quad (15)$$

The corresponding weights for these basis functions ($W_{1_e} \in \mathbb{R}, W_{2_e} \in \mathbb{R}$) are updated using the weight update rule given in Eq. (9).

5. SIMULATION AND RESULTS

In order to demonstrate the neuroadaptive learning scheme, the Mixed Meal tolerance test was simulated using the FDA-approved T1DM simulator for virtual patients. The perturbations due to exercise were added to the nominal model as per Eq.(2). The set of identified parameters ($\hat{\theta}_B$) for the MMT is for the breakfast cycle. The plant

model with a nominal set of parameters was simulated for a three-meal cycle where three meals and three corresponding insulin boluses were administered to the virtual patient. The simulation starts at 08:00 hours. The details of the meal time, meal amount, and insulin dose are given below in Table 1.

Table 1. Meal Cycle and Insulin Doses

Meal Time (hours)	Amount (grams)	Dose Time (hours)	Bolus Time (Units)
8:20	80	8:10	8
13:20	100	13:10	11
19:00	70	18:50	9

In order to account for parameter variations during lunch and dinner time, uncertainties were introduced in the Glucose and Remote Insulin state parameters as highlighted in Eq. (10). The true parameters are perturbed up to 40% of the known parameter value ($\Delta p_1, \Delta k, \Delta S_I$). In addition to the parameter uncertainty during different meals, the effect of exercise has also been incorporated into the simulation by varying the glucose effectiveness and insulin sensitivity (p_1, S_I) as highlighted in Eq. (2). The details of the physical activity that was simulated are shown below:

Table 2. Physical Activity

Start Time (hours)	Duration (mins)	Exercise Intensity (between 0 and 1)
11:00	50	0.5
21:20	60	0.5

Therefore, accounting for these parameter uncertainties and exercise effect, a simulation is shown below in Fig.1. There are three virtual patient simulations (i) Patient with nominal parameters, where all three meals use the same parameters θ_B , (given by blue dotted line) (ii) Patient with actual parameters, where a different set of parameters are used for Breakfast, Lunch and Dinner (given by red dashed line), and (iii) Patient with actual parameters and the exercise effect incorporated as well (given by green solid line). The purple vertical lines represent the amount of insulin bolus given to the patient (magnitude given on the right side of the y-axis). The meal intake period is shaded in blue, while the exercise period is shaded in red in Fig. 1. Therefore, it can be seen that the identification of just one set of parameters leads to significant errors in the plant model simulations. Utilizing this minimal model with only one set of parameters for control algorithm computation will lead to administering incorrect insulin doses. Moreover, it is seen that the effect of exercise increases glucose degradation, which can lead to hypoglycemia.

The adaptation of the model (with weights and basis functions) results in a more accurate patient simulation. The Root Mean Squared Error (RMSE) between the virtual patient model simulation with only nominal parameters (given by a blue solid line in Fig.2) and the patient model with varied parameters and exercise effect (given by green dashed line) is 7.33 mg/dL. The RMSE between the nominal patient with neuroadaptive learning and the actual patient is 1.13 mg/dL. This shows that the Neuroadaptive scheme can improve a nominal model parameter such that it mimics the dynamics of an actual patient.

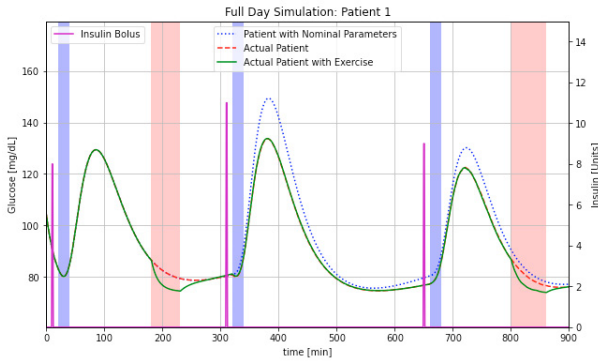


Fig. 1. Parameter Uncertainty during Lunch and Dinner and Exercise Effect.

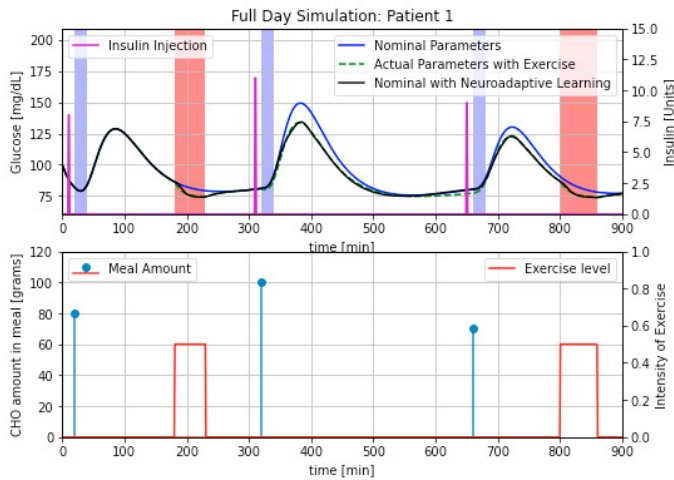


Fig. 2. Neuroadaptive Learning to adapt to Parameter Uncertainty and Exercise Effect.

6. CONCLUSION

In this work, a neuroadaptive learning technique is proposed in order to account for variations in the glucose-insulin dynamics and the effect of exercise. This scheme can be used in model-based control techniques such as Model Predictive Control and Back-stepping based control for Artificial Pancreas systems. These techniques rely heavily on a mathematical model of the patient in order to compute insulin doses necessary for blood glucose regulation. The adaptive scheme presented here can significantly improve the control accuracy and can compensate for the exercise effect. Once the effect of exercise is known, the insulin dose can be adjusted accordingly to prevent the risk of hypoglycemia. The future scope of this work is to augment this technique with the control scheme in an Artificial Pancreas system that has already been developed by the authors.

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